Welcome to Mars. Please Don’t Die.

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Description

A distributed automation platform capable of ingesting heterogeneous sensor data, normalizing it into a **unified internal representation**, evaluating simple **automation rules** dynamically to allow both manual and automatic activation of actuators, and providing a **real-time dashboard** for habitat monitoring and **plots** showing recent history of data.

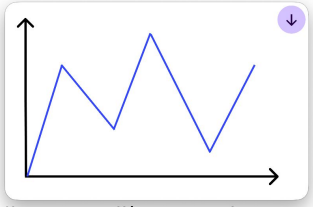
User Stories & Mockups

1. As a Mars operator, I want to visualize current data coming from REST sensors so that I can monitor the situation.
2. As a Mars operator, I want to visualize telemetries coming from sensors so that I can monitor the situation.
3. As a Mars operator, I want to be able to select the specific sensor data I want to plot so that I can analyze them in detail.
4. As a Mars operator, I want to be able to manually refresh values captured by the sensors so that I can get the most up to date information.
5. As a Mars operator, I want to set the value of the refreshing interval so that I can decide the frequency.
6. As a Mars operator, I want to add an automation rule to automate my reaction to the environment.
7. As a Mars operator, I want to update the threshold of an automation rule so that I can customize my rules.
8. As a Mars operator, I want to delete an automation rule so that I can get rid of it if I don't need it anymore.
9. As a Mars operator, I want to be able to visualize the actual state of all the actuators so that I can monitor the situation.
10. As a Mars operator, I want to manually change the state of the actuators so that I can have more control.
11. As a Mars operator, I want the values over the threshold to be highlighted so that I can easily understand what is happening.
12. As a Mars operator, I want to see some metrics (average, max, min) about received data so that I can analyse them.
13. As a Mars operator, I want to see the last triggered automation rule so that I know when the system reacts automatically.
14. As a Mars operator, I want to be able to export useful data into pdf files so that I can process them offline.
15. As a Mars operator, I want to be able to activate/disable an automation rule so that I can have more flexibility and handle critical situations.
16. As a Mars operator, I want to be able to see the list of automation rules, so that I can manage them.

MARS PROJECT | DASHBOARD | AUTOMATION RULES | ACTUATORS

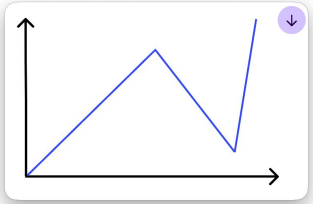
SENSOR NAME

Metric 1



Max: Min: Avg:

Metric 2



Max: Min: Avg:

MARS PROJECT | DASHBOARD | AUTOMATION RULES | ACTUATORS

ACTUATOR 1
ON OFF

ACTUATOR 2
ON OFF

ACTUATOR 3
ON OFF

ACTUATOR 4
ON OFF

MARS PROJECT | DASHBOARD | AUTOMATION RULES | ACTUATORS

AUTOMATION RULE #1

Sensor:
Operator:
Value:
Actuator:
ON|OFF:

Change threshold

DELETE

DISABLE

ADD A NEW RULE

AUTOMATION RULE #2

Sensor:
Operator:
Value:
Actuator:
ON|OFF:

Change threshold

DELETE

ENABLE

MARS PROJECT | DASHBOARD | AUTOMATION RULES | ACTUATORS

RULE TRIGGERED:

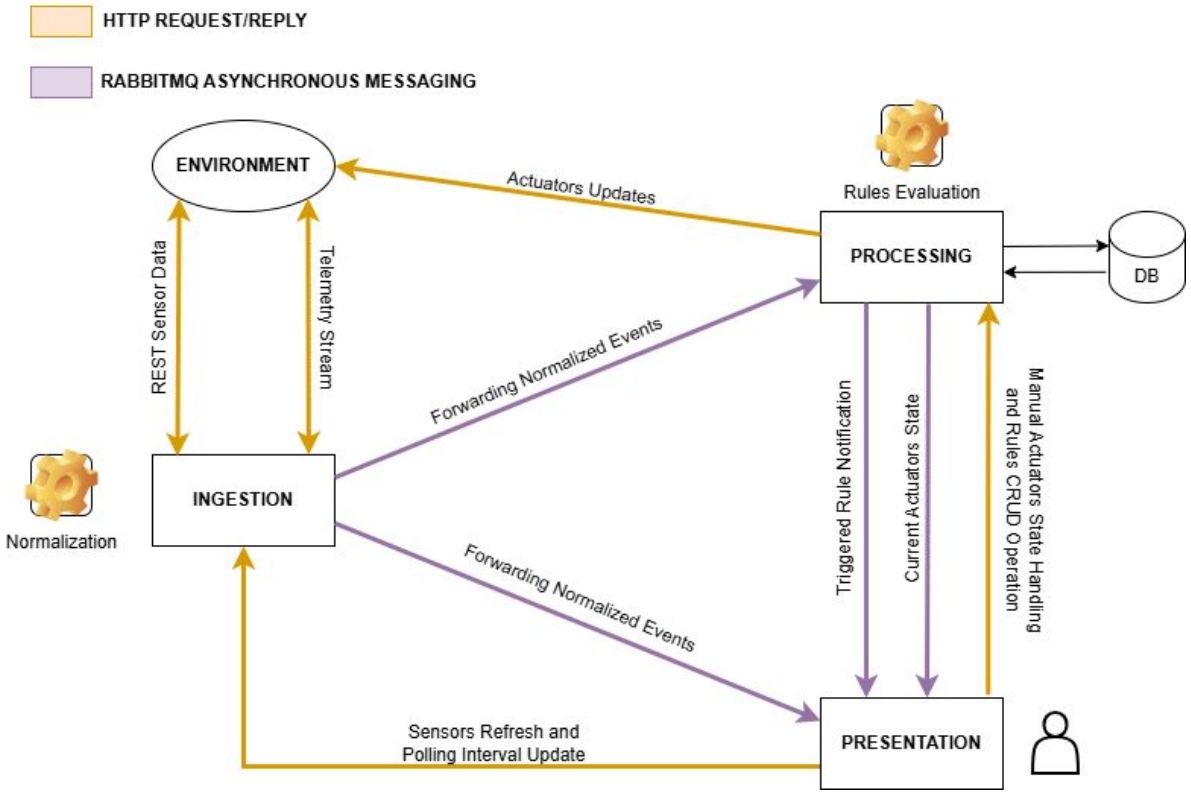
IF <sensor_name> <operator> <value> [unit]
THEN set <actuator_name> to ON | OFF

Polling every 0000 ms

REST SENSOR
Metric 1
Metric 2

TELEMETRY STREAM
Metric 1
Metric 2

Architectural diagram



Unified Event Schema

```
public class NormalizedEvent {
    private String sourceId;
    private SourceType sourceType;
    private String schema;
    private String timestamp;
    private String status;
    private List<Measurement> measurements;
…
}
```

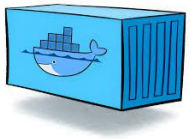
```
public class Measurement {
    private String metric;
    private Double value;
    private String unit;
…
}
```

```
public enum SourceType {
    REST,
    STREAM
}
```

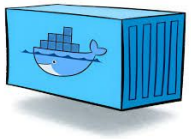
Technologies used

- Docker
- Spring Boot
- RabbitMQ
- PostgreSQL
- HTML, CSS, JavaScript

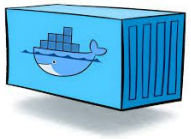
Containers



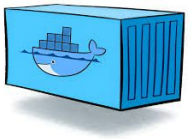
simulator



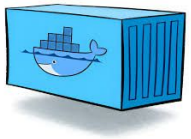
ingestion



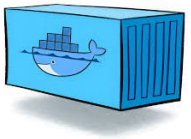
postgres



processing



rabbitmq



presentation

It’s demo time!

Thank you for your attention!